

Intelligent Complaint Management & Resolution Agent

An AI-Powered Customer Complaint Handling System

Project Overview

The Intelligent Complaint Management & Resolution Agent is an AI-powered system designed to handle customer complaints in a structured and efficient manner. The system understands the customer's complaint, identifies and classifies the issue, determines the appropriate next action, uses the required information and tools, and finally provides a suitable response to the customer. The overall objective is to make complaint handling faster, more organized, and more consistent while allowing complex cases to be transferred to human support when necessary.

System Workflow

The complete complaint-handling process follows a simple five-stage workflow: **Understand** → **Classify** → **Decide** → **Act** → **Respond**. First, the system understands the customer's complaint and extracts the relevant information. It then classifies the complaint according to its issue or category. After classification, the AI decides the appropriate action that should be taken. The system then performs the required action using available tools or APIs. Finally, it generates a structured response and communicates the outcome to the customer.

Main Features

The system provides automated complaint handling so that common customer issues can be processed without unnecessary manual intervention. It supports complaint classification and complaint status tracking, helping both customers and support teams understand the current state of an issue. AI-based decision-making is used to determine suitable next steps. Tool and API integration allows the system to obtain required information or perform relevant operations. For complex or exceptional cases, the system supports human escalation. The final customer response is presented in a structured and understandable form.

Example Scenario

Consider a customer who reports: **"My payment was deducted, but my order is not confirmed."** The system can identify this as a payment-related complaint and place the complaint in an **Under Review** status while the required information is checked and the appropriate resolution process is followed. This example demonstrates how the system can convert an unstructured customer message into a categorized and trackable complaint.

Proposed Technology

The proposed frontend can be developed using HTML, CSS, and JavaScript, or by using React for a more structured and interactive interface. The backend can be implemented in Python using FastAPI or Flask. The AI layer can use a Large Language Model (LLM), Prompt Engineering, and an Agentic AI Workflow to understand complaints and determine suitable actions. Complaint and customer data can be stored using MySQL or MongoDB. REST APIs and external tools can be integrated to support information retrieval and automated actions.

Future Scope

The system can be extended with voice-based complaint handling and multilingual support so that customers can communicate using different languages and interaction methods. WhatsApp and chat integration can provide additional communication channels. Sentiment analysis can help identify the emotional context of complaints. An analytics dashboard can provide insights into complaint patterns and resolution activity. CRM integration can connect complaint management with existing customer-support systems, while automated notifications can keep customers informed about complaint status and updates.

Team

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“Understand. Decide. Act. Resolve.”